Sabrina Sultana

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LinkedIn GitHub

Research Interests

- Efficient and Hardware-Aware Machine Learning and Algorithm–Hardware Co-Design for Edge AI Inference
- ML-Guided Design Space Exploration, Performance Modeling, and Mapping of Tensor Workloads on AI Accelerators and Parallel Hardware
- Efficient Adaptation and Inference of Large Models: Parameter-Efficient Fine-Tuning and Quantization
- Digital and VLSI Design for Edge AI Hardware

Education

Bachelor of Science in Electronics and Telecommunication Engineering
Chittagong University of Engineering and Technology (CUET) 2020–2025

- CGPA: 3.78/4.00; merit position 11th among 58 students
- English Proficiency: IELTS Academic, Overall Band 7.0
- Thesis: "Multi-Modal Deep Learning for Histopathological Stroke Diagnosis" (Supervisor: Priyanti Paul Tumpa, Assistant Professor, Dept. of ETE, CUET)
- Relevant Coursework: VLSI Technology, Digital Integrated Circuits, Microprocessor and Microcontroller, Digital Logic Design, Electronics-II, Semiconductor Physics and Devices, Data Structures and Algorithms, Object Oriented Programming, Digital Signal Processing

Research Publications

Conference Papers

- **Surrogate Modeling for ViT–Accelerator Co-Design: Generalization and Design-Space Exploration.** Under Review
  Sabrina Sultana, Anik Paul.
  ICREST 2027 (5th Int. Conf. on Robotics, Electrical and Signal Processing Techniques), AIUB, Dhaka.
  Contribution: GraphSAGE surrogate predicting cycle count, utilization, and mapping efficiency over a 12,096-point SCALE-Sim sweep (7.05% cycle-count MAPE on unseen architectures); limited to dense, single-core designs.
- **FedEdge: Quota-Guaranteed Tier Fairness for Hardware-Aware Federated Client Selection.** Under Review
  Sabrina Sultana, Mehlaqa, Intesur Chowdhury Ifsan, Yusuf Salehin.
  ICREST 2027 (5th Int. Conf. on Robotics, Electrical and Signal Processing Techniques), AIUB, Dhaka.
  Contribution: client-selection policy with a closed-form worst-case participation-fairness guarantee (under stated conditions), evaluated on real FedScale device traces.
- **Quantizing LoRA-Adapted FLAN-T5 for Dialogue Summarization: Quality, Faithfulness, and Efficiency Trade-offs.** Under Review
  Sabrina Sultana, Madhubi Omi, Anik Paul.
  ICEEICT 2027 (7th Int. Conf. on Electrical Eng. and Information & Communication Technology), MIST, Dhaka.
  Contribution: showed that aggregate scores stay flat under quantization while individual outputs change (54% rewritten under 4-bit).
- **Rethinking Decision Thresholds for TabPFN in Semiconductor Fault Detection.** Under Review
  Sabrina Sultana, Nur Alam, Iffat Ara, Yusuf Salehin.
  ICREST 2027 (5th Int. Conf. on Robotics, Electrical and Signal Processing Techniques), AIUB, Dhaka.
  Contribution: leakage-free evaluation with cross-validated threshold correction, calibration analysis (Brier score), and paired-bootstrap significance testing over 25 folds.
- **Seismic Risk Assessment in Chattogram: An Explainable Machine Learning Approach.** Presented
  Sabrina Sultana, Samia Shafique, Syeda Tamzida Akter, Choity Roy, Ayman Mahdia Khan, Md. Basir Zisan.
  iCONEECT 2026 (1st IEEE Int. Conf. on Next-Generation Electrical & Electronics, Computer Systems, and Technologies), Premier University, Chattogram.
- **BanglaLitEmo: A Source-Traceable Bangla Literary Emotion Dataset with LLaMA-3 Fine-Tuning for Fine-Grained Emotion Detection.** Under Review
  Nur Alam, **Sabrina Sultana**, Madhubi Omi, Taisirul Muktadi.
  ECCE 2027 (5th IEEE Int. Conf. on Electrical, Computer and Communication Engineering), CUET.

Journal Manuscript

- **Content-Aware Hybrid Multi-Model Fusion for Interpretable Histopathological Stroke Clot Classification.** Manuscript
  Sabrina Sultana, Priyanti Paul Tumpa.
  Dept. of Electronics and Telecommunication Engineering, CUET (undergraduate thesis work).
  Contribution: content-aware tile selection, fused Swin Transformer, MoCo and DINO embeddings, and attention heatmaps for interpretable whole-slide image classification.

Research Experience

Undergraduate Thesis Researcher
Supervisor: Priyanti Paul Tumpa Dec 2024 – June 2025
Dept. of Electronics & Telecommunication Engineering, CUET

- Investigated automated diagnosis of stroke from histopathological whole slide images (WSIs), a task with limited labeled data and no robust, generalizable models.
- Designed a hybrid Swin Transformer + self-supervised learning (MoCo, DINO) architecture producing 1024-dimensional fused feature representations; built a tile-selection and REMBG preprocessing pipeline to process 758 WSIs.
- Reduced 64 candidate tiles per slide to the 16 most informative via content-aware selection, and generated attention heatmaps for interpretability, reconstructed to full-slide resolution.
- Achieved 94.74% sensitivity and 91.38% specificity, a 25.82% improvement over the best previously reported result on this task.
- Tools: PyTorch, OpenSlide, scikit-learn, OpenCV, Transformer architectures.

Projects

SAP-1 Microprocessor Architecture Simulation (VLSI/Digital Design)

- Simulated an 8-bit microprocessor (ALU, registers, control unit, instruction cycle) in Logisim to study digital logic and control-signal generation.
- Tools Used: Logisim.

Arduino-Based Multi-Sensor Monitoring System (Embedded/IoT)

- Designed a real-time embedded monitoring system with gas, flame, IR, and ultrasonic sensors, implementing threshold-based alerting logic in embedded C.
- Tools Used: Arduino, Embedded C, gas/flame/IR/ultrasonic sensors.

Retrieval-Augmented Generation (RAG) for Financial Document Analysis

- Built a RAG pipeline (Llama-3, semantic chunking, hybrid vector search) achieving an 18% improvement in retrieval accuracy over baseline.
- Tools Used: Llama-3, Python, vector search frameworks.

Technical Skills

Hardware & Design Tools: SCALE-Sim, Cadence Virtuoso, ModelSim, Logisim, Arduino.
Programming Languages: Python, C, C++, MATLAB, SQL.
Libraries & Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, Keras, OpenCV, OpenSlide, SpaCy, NLTK, NumPy, Pandas, Matplotlib, Seaborn.
Other: Git, Docker, Jupyter/Colab, MS Excel, LaTeX, MS Word, MS PowerPoint.

Experience

AI Business Research Specialist
iSmartu Technology BD Limited, Dhaka Oct 2026–Present

- Build enterprise LLM agents with retrieval-augmented generation (RAG) to automate sales, finance, and supply-chain workflows.

Assistant Executive – Dealer Channel Development
Sales & Marketing, Bangladesh Steel Re-Rolling Mills Ltd. (BSRM) Nov 2025–Sep 2026

- Owned end-to-end data quality for company-wide transaction-level sales records, identifying data gaps, inconsistencies, and performance trends across customers, products, regions, salespersons, and B2B/B2C channels.
- Supported executive-level S&OP and senior management planning by transforming validated sales data into actionable performance analyses.

Telecommunications Engineer Intern
Teletalk Bangladesh Ltd., Chittagong 2024

- Gained hands-on exposure to mobile network technologies (1G–5G), network architecture, and communication protocols in a real-world telecommunications environment.
- Applied signal processing and network security concepts, including VPN configuration and firewall implementation, in telecommunications system planning.

Awards and Certifications

- Dean’s Award Nominee, CUET (ranked 11th of 58 in departmental merit list)
- Machine Learning Specialization, DeepLearning.AI
- Deep Learning Specialization, DeepLearning.AI
- Data Analytics Research Methods, SheStem & 10 Minute School